

Braeden West

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EDUCATION

Bachelor of Science in Computer Science (Software Engineering) Aug. 2024 – May 2026
Arizona State University *Tempe, AZ*

Associate of Arts in Computer Information Systems Aug. 2023 – Aug. 2024
Grand Rapids Community College *Grand Rapids, MI*

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Frontend Development, Backend Development, Databases & Storage, Computer & Network Security, Cloud Computing

EXPERIENCE

NASA Psyche Mission Sep. 2025 – May 2026
Technical Lead (Capstone) *Tempe, AZ*

- Led development of immersive 3D educational software for the NASA Psyche mission, now on permanent display at ASU's ISTB4 building, which hosts over 30,000 visitors annually.
- Collaborated with NASA scientists and university researchers to validate scientific accuracy and integrate domain expertise into interactive visualizations.
- Defined project scope, milestones, and team responsibilities, ensuring consistent alignment with sponsor requirements throughout the development lifecycle.

Jimmy John's Apr. 2023 – Aug. 2024
Delivery Driver *Rockford, MI*

- Delivered orders on tight time windows during high-volume shifts while maintaining strong customer service.
- Recognized as Employee of the Month multiple times for reliability and performance.

PROJECTS

Convolutional Neural Network | *Java, Git*

- Implemented a CNN from scratch in Java, manually building convolutional, pooling, and fully connected layers without any ML frameworks, including a custom Matrix class for all linear algebra operations.
- Achieved ~99% classification accuracy on the MNIST and Google Quick Draw datasets, validating end-to-end training, inference, and generalization.

UDP General Transfer System | *C++, CMake, OpenCV, Git*

- Built a real-time UDP transport layer from scratch in C++, implementing the full stack from raw sockets to a multithreaded producer-consumer pipeline for capture, transmission, and reassembly.
- Designed a fixed-width binary packet protocol and thread-safe queues to stream live video at 30fps with zero packet loss under burst traffic, demonstrating low-latency, loss-tolerant data transport suited to unreliable links.

Deterministic Finite Automaton Parser | *C++, Git*

- Built a custom lexer to tokenize formal language definitions, parsing states, alphabet, transition functions, start state, and accept states into a constructed automaton.
- Validated arbitrary input strings against the automaton and supported multiple DFA definitions and input sequences in a single run.

Project Spyn | *MATLAB, LEGO EV3*

- Designed and built an autonomous maze-navigating vehicle, integrating a 360° rotating ultrasonic sensor with real-time control logic for obstacle detection and pathfinding.
- Implemented dynamic environment mapping, building a live layout of the maze with checkpoint stops simulating waypoint navigation and hybrid manual/autonomous control modes.

TECHNICAL SKILLS

Programming Languages: Java, Kotlin, C#, C, C++, Swift, Python, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks & Libraries: React, Node.js, Three.js, Vite, Tailwind CSS, REST APIs, OAuth 2.0

Development Tools: Git, GitHub, Azure, Cloudflare, Docker, Unity, Blender, VSCode, IntelliJ, XCode